

Vishwa Patel

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Education

University of Southern California, CA, USA

Jan 2026–Dec 2027

- Master of Science (M.S.), Computer Science

GPA: 3.66/4

- Courses: Analysis of Algorithms, Web Technologies, Database Systems, Agentic AI, Natural Language Processing

Gujarat Technological University, Gujarat, India

Aug 2019–Jun 2023

- Bachelor of Engineering (B.E.), Computer Engineering

GPA: 8.98/10

- Courses: Data Structures, Object-Oriented Programming Principles, Theory of Computation, Compiler Design, Software Engineering, Machine Learning, Artificial Intelligence, Operating Systems

Technical Skills

- **Languages:** C++, Python, Java, C, C#, Go, JavaScript/TypeScript, SQL, HTML/CSS, Dart
- **Systems & Databases:** PostgreSQL (incl. pgvector), MongoDB, Redis, MySQL, SQLite, HBase, Cassandra, Database Internals, Query Optimization, Linux
- **Frameworks & Tools:** React, Node.js, FastAPI, REST APIs, Express, Flask, Next.js, Angular, GraphQL, Spring Boot, Spring MVC, Elasticsearch, RabbitMQ, jQuery, Flutter, GDB, GCC
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Firebase, Git, CI/CD, Microservices
- **AI/ML:** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, RAG, SpaCy, YOLOv11, Claude API, Voyage AI

Professional Experience

Casepoint Pvt. Ltd, Gujarat, India

Software Engineer

May 2025–Jun 2025

- Engineered custom field selection for the legal hold reporting module so clients could build personalized reports, cutting report generation time by 30% while staying within regulatory compliance.
- Resolved 10-20 high-priority support tickets a week for extraction and processing failures, diagnosing root causes and shipping code-level fixes to stay within client SLAs.
- Improved status visibility 90% by building a companion agent that tracks audio-peak collection in real time by request ID.

Associate Software Engineer

Jul 2023–May 2025

- Rebuilt the extraction workflow to run in parallel instead of single-threaded, boosting throughput by ~90%.
- Developed a PST repair service with automated recovery and retry logic for corrupted mailbox files, trimming manual intervention by 50 to 70% and integrating it directly into the production workflow.
- Leveraged Redis to track document status across 6+ workflow stages, staging each record during ingestion and committing it to the database only once processing completes, slashing status-update latency by 80%.
- Extended the metadata service to support a new file format for forensic analysis, folding the update into the same pipeline.
- Upgraded core extraction libraries (Oracle, Aspose, VintaSoft) for better compatibility, reducing errors and failures by over 40%.
- Partnered with 5+ teams across product and engineering to scope requirements and ship 8+ production features, including surfacing operational data to other teams' dashboards without breaking integrations.

Software Engineer Intern

Jan 2023–Jun 2023

- Created a desktop-based Recruitment System (.NET MVC, C#, Windows Forms, PostgreSQL, Kendo UI) with 20+ UI screens and both client- and server-side validation for better maintainability.
- Integrated RabbitMQ for asynchronous messaging between app components as part of an Agile sprint cycle.

Amiseq, Pune, India

Software Engineering Intern (Backend)

Jun 2022–Jul 2022

- Shipped 10+ RESTful API endpoints in Spring Boot and PostgreSQL, structured as modular microservices for CRUD operations.
- Debugged and validated those endpoints in Postman, catching integration issues early and confirming reliable request/response handling.

Projects

GTM Signal Agent | Python, FastAPI, React, PostgreSQL | [GitHub](#)

- Architected an AI sales agent (FastAPI, PostgreSQL) that turns company funding, hiring, and tech-stack signals into outreach briefs, with a model layer spanning three backends: Groq, Ollama, and Anthropic.
- Introduced a risk-scoring engine that flags stale deals, so reps know within a rolling 30-day window which accounts need follow-up.

Smart Document Reader | Python, FastAPI, PostgreSQL/pgvector, Ollama | [GitHub](#)

- Designed a local-first RAG assistant that combines format-aware extraction with hybrid dense/lexical retrieval to parse PDFs and tables accurately, grounding every answer with a citation back to its exact source.
- Benchmarked retrieval accuracy across 16 configurations to identify the best-performing setup.

TechHire: AI Job Search Platform | Python, FastAPI, React, PostgreSQL | [GitHub](#)

- Launched a full-stack job platform (200+ live postings) with filtering by skills, salary, visa sponsorship, and work mode; the search API averages ~7ms across the full index.
- Paired it with an AI resume analyzer (Groq/Llama, 3-model pipeline) that scores candidate fit 0 to 100 and rewrites weak bullets; in blind testing it showed an 85-point spread between strong- and weak-fit resumes.

Sequence Alignment | Python | [GitHub](#)

- Implemented global DNA sequence alignment with Needleman-Wunsch DP and Hirschberg's divide-and-conquer, shrinking space complexity from $O(mn)$ to $O(m+n)$ with no change in the optimal alignment output.
- Measured CPU time and peak memory usage across different sequence lengths and checked correctness against a brute-force baseline.

Certifications & Awards

- Python for Everybody Specialization (Coursera): Certification in Python, databases, web scraping, and data visualization.
- Casepoint Excellence Award 2024 (Sprint Delivery Standout): Awarded for outstanding sprint delivery and performance.